Levels of Transparencies in DDBMS

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What is Transparency?

- In a Distributed Database (DDBMS), data is kept in many different computers.
- **Transparency** means the user does not see this complexity.
- For a student or a user, it feels like using just one single database on one computer.
- It hides the "where" and "how" of the data.

1. Data Distribution Transparency

This level hides how the data is physically spread out.

- **Location Transparency:** You don't need to know the city or server name where data is stored.
- **Fragmentation Transparency:** If a big table is cut into small pieces (fragments), you don't need to worry. You just use the table name.
- **Replication Transparency:** If the system keeps many copies of data for safety, you only see one.

2. Transaction Transparency

A "Transaction" is any job you do in the database (like a bank transfer).

- **Concurrency Transparency:** Many people can use the database at the same time without breaking things.
- **Failure Transparency:** If one computer or network cable fails, the database ensures your work is either finished completely or cancelled safely.

3. DBMS Transparency (Heterogeneity)

- Sometimes, one office uses *MySQL* and another uses *Oracle*.
- **DBMS Transparency** hides these differences.
- The user writes one simple query, and the system translates it for different database softwares automatically.

The Transparency Layers

Mathematically, the system follows this flow to keep things clear for the user:

$$\text{User View} \rightarrow \text{Global Schema}$$

$$\downarrow$$

$$\text{Fragmentation Schema} \rightarrow \text{Allocation Schema}$$

$$\downarrow$$

$$\text{Local Physical Database}$$

Conclusion

- Transparency is the most important feature of a DDBMS.
- It makes a complicated network of computers look like a simple, single folder.
- It helps us work faster without worrying about network errors or data locations.

References

1. M. Tamer Özsu and Patrick Valduriez, *Principles of Distributed Database Systems*.
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3. C.J. Date, *An Introduction to Database Systems*.